

DFT studies of Al–O Raman vibrational frequencies for aquated aluminium species

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ABSTRACT

Raman frequency calculations of $\text{Al}(\text{H}_2\text{O})_6^{3+}$ and $\text{Al}(\text{OH})_4^-$ species in different models were carried out with density functional theory (DFT). Gas-phase model (GP), polarizable continuum model (PCM), supermolecule model (SM) and supermolecule-polarizable continuum model (SM-PCM) were considered and frequencies over the wavenumber range from 0 to 1000 cm^{-1} were discussed. The calculated frequencies of the two species in SM-PCM model are in fair agreement with the observed frequencies in aqueous solution and the average deviation is 6 cm^{-1} . Based on the results of $\text{Al}(\text{H}_2\text{O})_6^{3+}$ and $\text{Al}(\text{OH})_4^-$, the DFT method and the SM-PCM model were used to study the Al–O Raman bands for some other monomeric and dimeric aluminiums. The monomeric aluminiums $\text{Al}(\text{OH})_2^+$, $\text{Al}(\text{OH})_2^+$ and $\text{Al}(\text{OH})_3^{3-}$ as well as the dimer $\text{Al}_2\text{O}(\text{OH})_6^{2-}$ were included. As a result, we predict the main Al–O skeleton vibrations of $\text{Al}(\text{OH})(\text{H}_2\text{O})_5^{2+}$ at 685 cm^{-1} , *trans*- $\text{Al}(\text{OH})_2(\text{H}_2\text{O})_4^+$ at 542 cm^{-1} , $\text{Al}(\text{OH})_3^{3-}$ at 510 cm^{-1} and $\text{Al}_2\text{O}(\text{OH})_6^{2-}$ at 510 and 699 cm^{-1} respectively in Raman spectra of aqueous solution. It can also be concluded that the calculated frequencies of $\text{Al}_2\text{O}(\text{OH})_6^{2-}$ are close to the experimental values and the existence of $\text{Al}(\text{OH})_3^{3-}$ in aluminate solution cannot be ruled out.

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1. Introduction

Aluminium species in aqueous solution have been extensively studied for years [1]. Some species have been identified, e.g. $\text{Al}(\text{H}_2\text{O})_6^{3+}$, $\text{Al}_2(\text{OH})_2(\text{H}_2\text{O})_8^{4+}$, $\varepsilon\text{-Al}_{13}\text{O}_4(\text{OH})_{24}(\text{H}_2\text{O})_{12}^{7+}$, $\text{Al}(\text{OH})_4^-$, $\text{Al}_2\text{O}(\text{OH})_6^{2-}$, etc. [2] some hydrolysis products, such as $\text{Al}(\text{OH})_2^+$ and $\text{Al}(\text{OH})_2^+$ are believed to exist [3] but their coordination numbers are still in dispute [4,5]. Meanwhile, there are some complexes whose existences are the focus of attention, including $\text{Al}(\text{OH})_5^{2-}$, $\text{Al}(\text{OH})_6^{3-}$, $\text{Al}_2(\text{OH})_8^{2-}$, etc. [1a,6,7].

Raman spectrum is ideal for the studies of aluminium(III) species in aqueous solution since water is a weak Raman scatter and band assignments can be facilitated by isotopic substitution and polarization spectra [8]. Based on the results of Raman spectra, $\text{Al}(\text{H}_2\text{O})_6^{3+}$ has been thought to be thermodynamically stable in chloride, nitrate and perchlorate solutions and the hydrolysis of aluminium(III) aqua ion can be observed in sulfate and phosphate solutions [9,10]. Meanwhile, Raman spectra have been used to study the possible structures and mutual transformation in aluminate solution [1a,7,11].

Few research has been done to simulate Al–O vibrational frequencies of aquated aluminium species. $\text{Al}(\text{H}_2\text{O})_6^{3+}$ and $\text{Al}(\text{OH})_4^-$ are the main calculated complexes and the models including the

second hydration sphere are believed to be conducive to frequency calculations for the two clusters [9b,12]. By the calculation of reaction energies and frequencies of the possible species, Gale et al. proposed that $\text{Al}(\text{OH})_4^-$ and $\text{Al}_2(\text{OH})_8^{2-}$ occur in the dilute and concentrated aluminate solutions respectively [6b]. In this study, we try to establish an accurate model for Al–O Raman frequency calculations in aqueous solution and then use these conditions to predict Al–O Raman vibrational values for the monomeric and dimeric aquated aluminium species concerned by people. It provides a reference to determine the existence of some species in aqueous solution of aluminium.

2. Computational details

Four models were used including: (1) gas-phase model (GP), in which species are fully optimized in vacuum; (2) polarizable continuum model (PCM), in which the bulk solvent effect is considered on the gas-phase species [13]; (3) supermolecule model (SM), in which 12 water molecules are added as a complete second hydration shell with the consideration of the explicit solvent effect [5e,14]; (4) supermolecule-polarizable continuum model (SM-PCM), in which the bulk solvent effect is considered on the supermolecules [5,15]. It must be mentioned that the second hydration shells were established based on the similar structures from previous Refs. [14,16]. The arrangements achieved the first hydration shells with total hydrogen bond and some of our previous work has proved these structures to be reliable [5d,5e].

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All calculations were carried out with the Gaussian 03 computer program [17]. All structures were optimized to be local minima at B3LYP/6-311 + G(d,p) without any symmetry constraints [18]. The structures in GP and SM models were optimized in vacuum, while in PCM calculations the dielectric constant 78.39 for water at 298.15 K and the UAKS radii were used to account for the bulk solvent effect [5]. In the article, the calculated frequencies of $\text{Al}(\text{H}_2\text{O})_6^{3+}$ and $\text{Al}(\text{OH})_4^-$ in the previous four models were discussed firstly and 12 water molecules were included in the second hydration shell (Fig. 1). Then the model in which the calculated results were closest to the observed frequencies was used for the monomeric and dimeric aquated aluminium species including $\text{Al}(\text{OH})_2^{2+}$, $\text{Al}(\text{OH})_2^+$, $\text{Al}(\text{OH})_3$ and $\text{Al}_2\text{O}(\text{OH})_6^{2-}$. The isomers were also calculated as follows: (1) $\text{Al}(\text{OH})(\text{H}_2\text{O})_5^{2+}$ and $\text{Al}(\text{OH})(\text{H}_2\text{O})_4^+$ for $\text{Al}(\text{OH})_2^{2+}$; (2) *cis*- $\text{Al}(\text{OH})_2(\text{H}_2\text{O})_4^+$, *trans*- $\text{Al}(\text{OH})_2(\text{H}_2\text{O})_4^+$, *cis*- $\text{Al}(\text{OH})_2(\text{H}_2\text{O})_3$, *trans*- $\text{Al}(\text{OH})_2(\text{H}_2\text{O})_3$ and $\text{Al}(\text{OH})_2(\text{H}_2\text{O})_2^+$ for $\text{Al}(\text{OH})_2^+$. The distribution of the second hydration layer for all the complexes were shown in Fig. 2.

The calculations of Raman vibrational frequencies and natural population analysis [19] were employed on the optimized structures at the same theoretical level of B3LYP/6-311 + G(d,p). The low-symmetry structures in our calculation were agreement with the equilibrium configuration of aqua metal ion in solution [20], which shows our research to be of universality. The main Raman Al–O vibrations for all structures were assigned by the assistant of normal coordinate analysis and animated motions with the GaussView program. We are interested in the vibrations of Al–O which always occur at wavenumbers below 1000 cm^{-1} [1a,7,9–11,21]. Therefore, only the calculated values in the range 0– 1000 cm^{-1} were discussed. The calculated harmonic frequencies should have been scaled to correct for the approximate treatment of electron correlation, basis set deficiencies and anharmonicity [22]. The predicted frequencies of metal–oxygen (M–O) of aquated cations always underestimate the experimental values [23]. The scaling factors for vibrations of organic molecules do not fit the M–O vibrations. Hence, we have just simply reported the unscaled harmonic frequencies. As results of low structural symmetry and coupling with the modes of H_2O , the same Al–O vibration was always observed at two or more frequencies [23e,23f]. To compare

with the experimental value, we use their average as the calculated vibrational frequency [23e,23f].

3. Results and discussion

3.1. $\text{Al}(\text{H}_2\text{O})_6^{3+}$ and $\text{Al}(\text{OH})_4^-$

3.1.1. Structures of $\text{Al}(\text{H}_2\text{O})_6^{3+}$ and $\text{Al}(\text{OH})_4^-$

The Al–O bond length and charge distribution are the main factors to determine vibrational frequencies [8a,24]. The structural parameters of $\text{Al}(\text{H}_2\text{O})_6^{3+}$ and $\text{Al}(\text{OH})_4^-$ in GP, PCM, SM and SM-PCM models are listed in Table 1. The calculated frequencies vary with solvation models where the interactions between solvent and solute are considered differently. For $\text{Al}(\text{H}_2\text{O})_6^{3+}$, the Al–O bond length from GP model is significantly longer than the experimental value as a result of the neglect of the solvent–solute interactions. However, a certain degree of improvement can be observed from the results of all the other models and the SM and SM-PCM models give the similar structure data. For $\text{Al}(\text{OH})_4^-$, the Al–O bond lengths in different models remain almost unchanged except that the value from PCM model is shorter, which shows the PCM model neglecting direct chemical action such as hydrogen bonds and charge transfer is improper [13d,25]. Meanwhile the results of $\text{Al}(\text{OH})_4^-$ in SM and SM-PCM models are close to each other.

3.1.2. Comparison of the models for $\text{Al}(\text{H}_2\text{O})_6^{3+}$ Raman calculation

The Raman spectrum of the hexaaquaaluminium complex ion is characterized by three bands assigned to Al–O modes [10,21,23]. The bands at 525 and 332 cm^{-1} are ascribed to AlO_6 symmetric stretching mode ($\nu_s \text{AlO}_6$) and scissoring mode ($\delta \text{O–Al–O}$) respectively. The band at 438 cm^{-1} is assigned to symmetric stretch of O–Al–O with octahedron distorted and it is denoted as $\nu_{as} \text{AlO}_6$. The schematic diagrams of the modes are shown in Fig. 3. Fig. 4 depicts the calculated Raman spectra of $\text{Al}(\text{H}_2\text{O})_6^{3+}$ species in different models. Calculated Raman activities are not directly comparable to measured intensities [17,26] and they are not discussed in the article. Besides the three bands mentioned above, the vibrations of water molecules and other Al–O vibrational modes with low Ra-

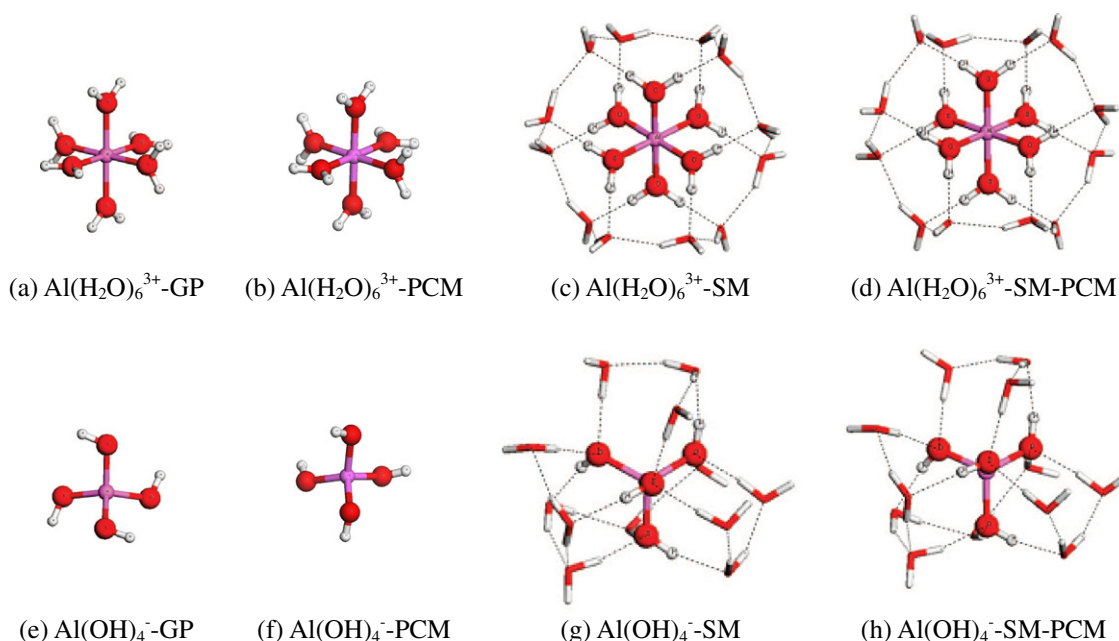


Fig. 1. The optimized structures of $\text{Al}(\text{H}_2\text{O})_6^{3+}$ and $\text{Al}(\text{OH})_4^-$ species in GP, PCM, SM, and SM-PCM models.

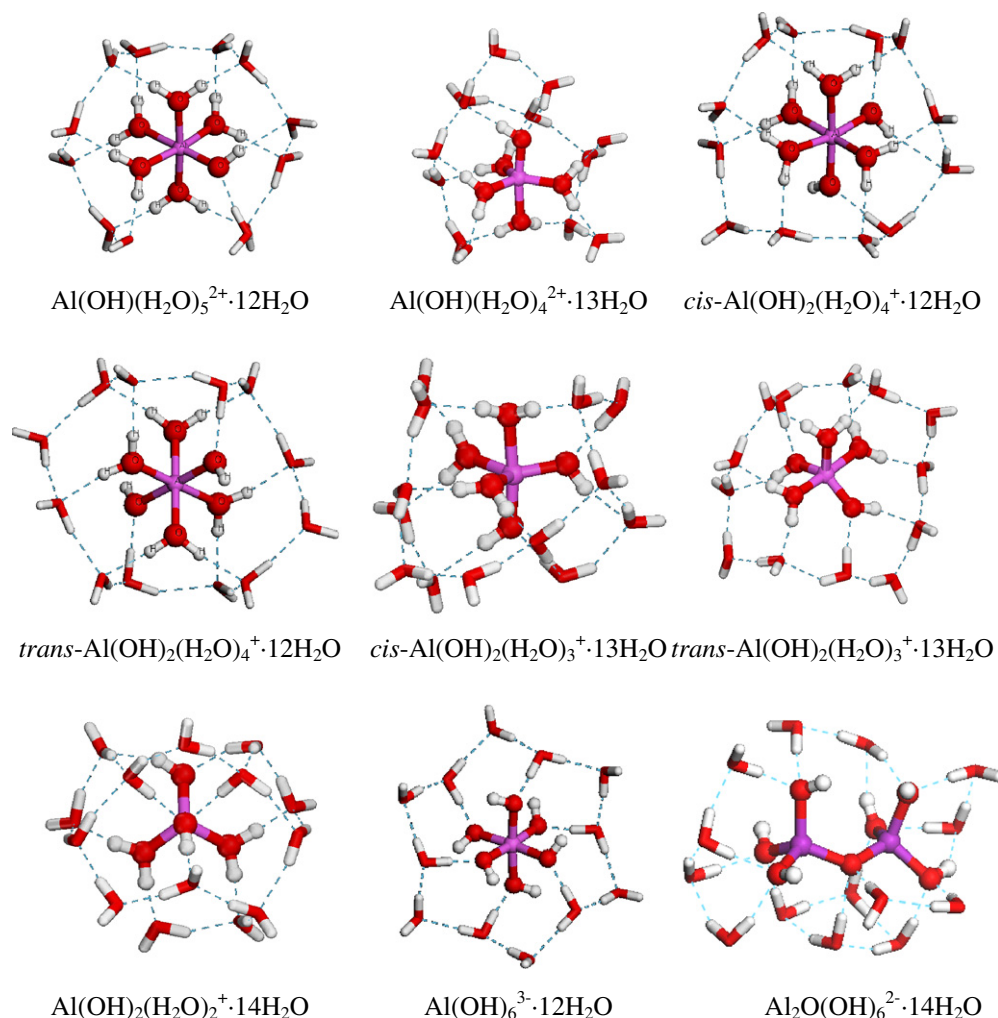


Fig. 2. The optimized structures of the monomeric and dimeric aquated aluminium species in SM-PCM model.

Table 1

Structural parameters and Al–O Raman frequencies of $\text{Al(H}_2\text{O)}_6^{3+}$ and Al(OH)_4^- in different models.

		GP	PCM	SM	SM-PCM	Exp. data
$\text{Al(H}_2\text{O)}_6^{3+}$	$B_{\text{Al-O}}/\text{\AA}^a$	1.941	1.899	1.921	1.913	1.90 ^d
	Q_{Al}/e^b	1.959	1.947	1.949	1.943	
	Q_{O}/e^c	−0.966	−0.972	−1.001	−1.002	
	$\delta \text{ O-Al-O}/\text{cm}^{-1}$	225	250	305	327	332 ^e
	$\nu_{\text{as}} \text{ AlO}_6/\text{cm}^{-1}$	347	395	408	426	438
	$\nu_{\text{s}} \text{ AlO}_6/\text{cm}^{-1}$	446	494	506	530	525
Al(OH)_4^-	$B_{\text{Al-O}}/\text{\AA}$	1.792	1.786	1.792	1.793	1.80 ^f
	Q_{Al}/e	2.033	2.043	2.087	2.087	
	Q_{O}/e	−1.207	−1.209	−1.250	−1.251	
	$\nu_{\text{s}} \text{ AlO}_4/\text{cm}^{-1}$	585	592	595	608	620 ^g

^a Bond length of Al–O (oxygen coordinated to aluminium).

^b Natural charge of Al.

^c Natural charge of O (coordinated to aluminium).

^d Al–O bond length from Ref. [23].

^e Raman data from Ref. [9,10].

^f Al–O bond length from Ref. [21].

^g Raman data from Ref. [1a,7,11].

man activities can also be observed in the calculated spectra. Clearly, the calculated Raman spectra become complicated as a result of the addition of the second hydration shell. The vibrational modes of water clusters can differ a lot in different models [27]. However, they are not our concern and have been classified as

“Others” in Fig. 4. The average Al–O frequency values are listed in Table 1. Obviously, the result of GP model has the largest deviation (92 cm^{-1} on average). In PCM model, the short Al–O bond length could give vibrations at higher wavenumbers. Unfortunately, the electrostatic interactions between the center aqua ion

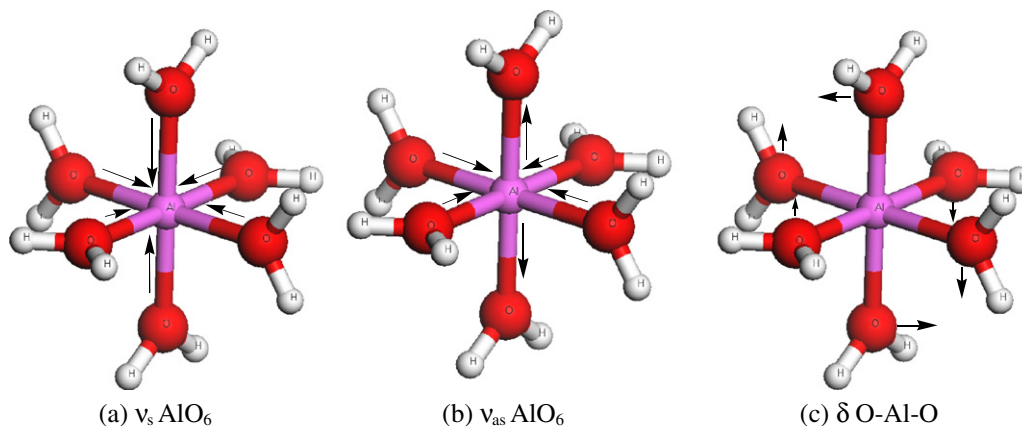


Fig. 3. The schematic diagrams of AlO_6 skeleton modes in $\text{Al}(\text{H}_2\text{O})_6^{3+}$.

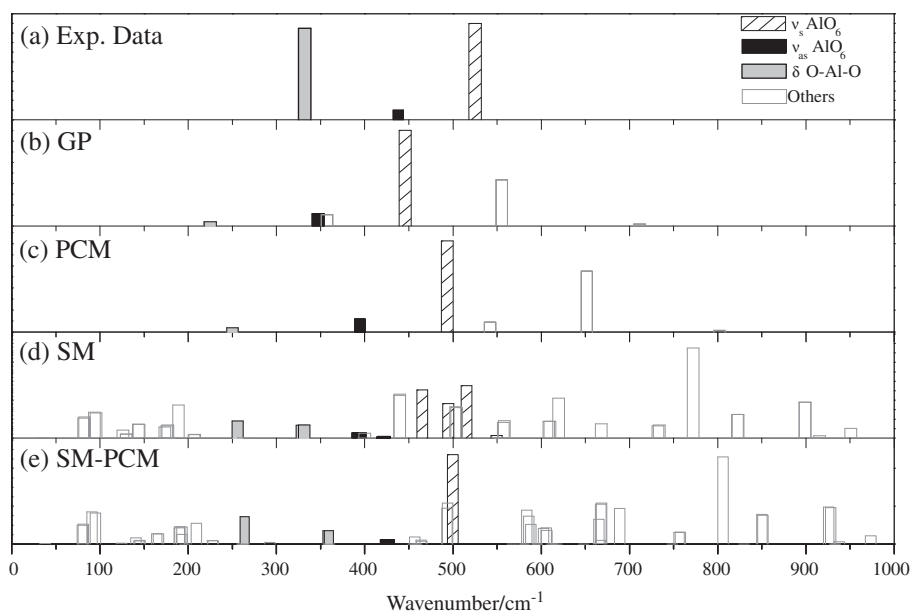


Fig. 4. Experimental and calculated Raman spectra of aluminium(III) hydration in solution. The experimental data is taken from Refs [4–6] and the modes of water clusters and anions are neglected. Others include the vibrations of water molecules and other Al–O vibrational modes with low Raman activities.

and the bulk solvent lower the electronic density of AlO_6 unit and weaken the Al–O bond strength. As a result, the calculated frequencies of AlO_6 are 52 cm^{-1} smaller than the observed values in aqueous solution. The results of SM species show significant improvement and the deviations of all modes are within 30 cm^{-1} . It is noteworthy that the results of SM-PCM model are in fair agreement with the experimental frequencies of aqueous $\text{Al}(\text{III})$ ion. The largest and average deviations are only 12 and 4 cm^{-1} , respectively.

3.1.3. Comparison of the models for $\text{Al}(\text{OH})_4^-$ Raman calculation

The main band in the Raman spectra of aluminate solutions at 620 cm^{-1} is assigned to AlO_4 symmetric stretching mode ($\nu_s \text{AlO}_4$, depicted in Fig. 5) [1a,7,11]. The calculated Raman spectra and average Al–O frequency values are shown in Fig. 6 and Table 1 respectively. Different models cause little difference in the values of $\nu_s \text{AlO}_4$. However, we can still observe that the change trends of frequencies are similar to those for $\text{Al}(\text{H}_2\text{O})_6^{3+}$. The calculation of $\text{Al}(\text{OH})_4^-$ in GP model gives the lowest value. The short bond distance of Al–O makes the vibration occur at a higher wavenumber. The electronic density and bond length from SM model are close to

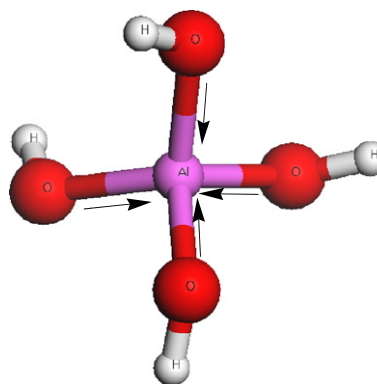


Fig. 5. The schematic diagram of $\nu_s \text{AlO}_4$ in $\text{Al}(\text{OH})_4^-$ species.

those from SM-PCM model and this can well explain the similar frequency values obtained from the two models. In the case of SM-PCM model, the calculated result (608 cm^{-1}) corresponds with the measured frequency of aluminate solution (620 cm^{-1}).

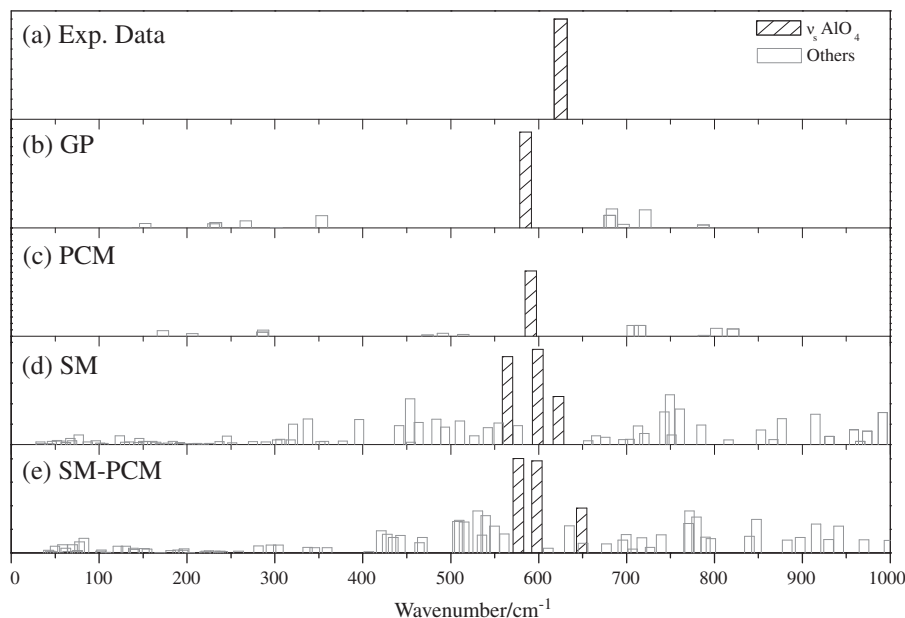


Fig. 6. Experimental and calculated Raman spectra of $\text{Al}(\text{OH})_4^-$. The experimental data is taken from Refs [1a,7,8] and the modes of water clusters and anions are neglected. Others include the vibrations of water molecules and other Al–O vibrational modes with low Raman activities.

3.2. Application to study on the monomeric and dimeric aquated aluminium species

The SM-PCM model can be used to well reproduce the Al–O Raman spectra in aqueous solution based on the previous discussion. So the model is used to discuss the Al–O Raman vibrations of other monomeric and dimeric aquated aluminium species. $\text{Al}(\text{OH})^{2+}$, $\text{Al}(\text{OH})_2^+$, $\text{Al}(\text{OH})_6^{3-}$, and $\text{Al}_2\text{O}(\text{OH})_6^{2-}$ in SM-PCM model were calculated and the results of structural parameters are listed in Table 2.

3.2.1. The monomeric aquated aluminium species

As the important hydrolysis products of $\text{Al}(\text{H}_2\text{O})_6^{3+}$, different coordination structures of $\text{Al}(\text{OH})^{2+}$ and $\text{Al}(\text{OH})_2^+$ were discussed. We modelled and calculated $\text{Al}(\text{OH})(\text{H}_2\text{O})_5^{2+}$ and $\text{Al}(\text{OH})(\text{H}_2\text{O})_4^+$ for the first hydrolysis product $\text{Al}(\text{OH})^{2+}$, *cis*- $\text{Al}(\text{OH})_2(\text{H}_2\text{O})_4^+$, *trans*- $\text{Al}(\text{OH})_2(\text{H}_2\text{O})_4^+$, *cis*- $\text{Al}(\text{OH})_2(\text{H}_2\text{O})_3^+$, *trans*- $\text{Al}(\text{OH})_2(\text{H}_2\text{O})_3^+$ and $\text{Al}(\text{OH})_2(\text{H}_2\text{O})_2^+$ for $\text{Al}(\text{OH})_2^+$. The structural diagrams and parameters are presented in Fig. 1 and Table 2. Obviously, six-coordination $\text{Al}(\text{OH})(\text{H}_2\text{O})_5^{2+}$ and *trans*- $\text{Al}(\text{OH})_2(\text{H}_2\text{O})_4^+$ are the most stable struc-

tures for $\text{Al}(\text{OH})^{2+}$ and $\text{Al}(\text{OH})_2^+$. Meanwhile it can be observed that the Al–O bond lengths in both Al–OH₂ and Al–OH increase with coordination number and hydrolysis degree. There are three kinds of Al–O vibrations with high Raman activities: (1) the stretching vibration of Al–OH₂ (ν Al–OH₂); (2) the stretching vibration of Al–OH (ν Al–OH); (3) the HO–Al–OH symmetric stretching vibration (ν_s HO–Al–OH) which only occurs in the structures with two hydroxyls in *trans*-position. Fig. 7 demonstrates the vibration modes with $\text{Al}(\text{OH})(\text{H}_2\text{O})_5^{2+}$ and *trans*- $\text{Al}(\text{OH})_2(\text{H}_2\text{O})_4^+$. The calculated frequencies of these structures are listed in Table 3. By comparison of the frequency values, we can conclude that the three kinds of Al–O vibration modes shift to low frequency with the increase of coordination and hydrolysis degree, which is consistent with the bond length changes. According to the result of $\text{Al}(\text{OH})(\text{H}_2\text{O})_5^{2+}$, ν Al–OH₂ and ν Al–OH appear at 476 and 685 cm^{-1} , respectively. The strongest mode of *trans*- $\text{Al}(\text{OH})_2(\text{H}_2\text{O})_4^+$ is at 542 cm^{-1} . Based on the previous discussion of $\text{Al}(\text{H}_2\text{O})_6^{3+}$ and $\text{Al}(\text{OH})_4^-$, the above-mentioned wavenumbers can be regarded as the main Raman frequencies of Al–O vibrations

Table 2

Structural parameters of the monomeric and dimeric aquated aluminium species in SM-PCM model.

Com. ^a	CN ^b	Mol. Form. ^c	E(Hartree) ^d	B(Å) ^e		Q(e) ^f		
				Al–OH ₂	Al–OH	Al	O(OH ₂)	O(OH)
$\text{Al}(\text{OH})^{2+}$	6	$\text{Al}(\text{OH})(\text{H}_2\text{O})_5^{2+} \cdot 12\text{H}_2\text{O}$	−1617.580956	1.953	1.803	1.943	−0.988	−1.209
	5	$\text{Al}(\text{OH})(\text{H}_2\text{O})_4^{2+} \cdot 13\text{H}_2\text{O}$	−1617.544918	1.892	1.780	2.034	−1.016	−1.220
$\text{Al}(\text{OH})_2^+$	6	<i>cis</i> - $\text{Al}(\text{OH})_2(\text{H}_2\text{O})_4^+ \cdot 12\text{H}_2\text{O}$	−1617.275128	1.993	1.825	1.939	−0.973	−1.208
	6	<i>trans</i> - $\text{Al}(\text{OH})_2(\text{H}_2\text{O})_4^+ \cdot 12\text{H}_2\text{O}$	−1617.284101	1.987	1.826	1.941	−0.975	−1.201
	5	<i>cis</i> - $\text{Al}(\text{OH})_2(\text{H}_2\text{O})_3^+ \cdot 13\text{H}_2\text{O}$	−1617.267820	1.942	1.788	2.021	−1.000	−1.213
	5	<i>trans</i> - $\text{Al}(\text{OH})_2(\text{H}_2\text{O})_3^+ \cdot 13\text{H}_2\text{O}$	−1617.273746	1.952	1.770	2.019	−0.994	−1.214
	4	$\text{Al}(\text{OH})_2(\text{H}_2\text{O})_2^+ \cdot 14\text{H}_2\text{O}$	−1617.267863	1.833	1.746	2.088	−1.045	−1.233
$\text{Al}(\text{OH})_6^{3-}$	6	$\text{Al}(\text{OH})_6^{3-} \cdot 12\text{H}_2\text{O}$	−1614.987448	–	1.952	1.945	–	−1.166
Al_2O^{2-}	4	$\text{Al}_2\text{O}(\text{OH})_6^{2-} \cdot 14\text{H}_2\text{O}$	−2086.5089335	–	1.800	2.097	–	−1.220

^a Compounds.

^b Coordinate number.

^c Molecular formulate.

^d Energy.

^e Bond length.

^f Natural charge.

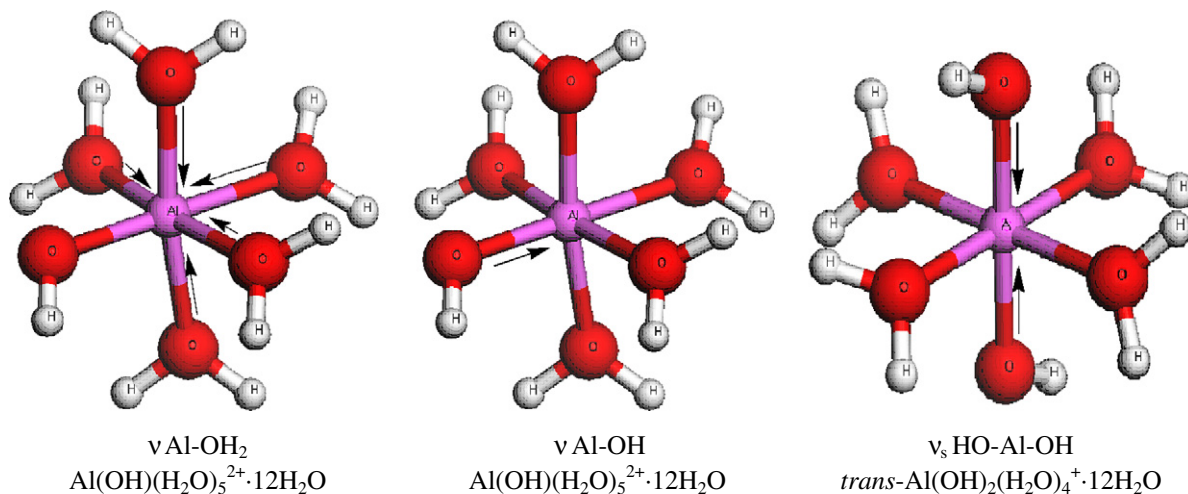


Fig. 7. The schematic diagrams of $\nu \text{ Al-OH}_2$, $\nu \text{ Al-OH}$ and $\nu_s \text{ HO-Al-OH}$ in the hydrolysis products of $\text{Al(H}_2\text{O)}_6^{3+}$ (Al(OH)^{2+} and Al(OH)_2^+).

Table 3

The calculated frequencies of Al–O Raman vibrations of Al(OH)^{2+} and Al(OH)_2^+ (cm^{-1}).

Al species	$\nu \text{ Al-OH}_2$	$\nu_s \text{ HO-Al-OH}$	$\nu \text{ Al-OH}$
$\text{Al(OH)(H}_2\text{O)}_5^{2+} \cdot 12\text{H}_2\text{O}$	479	–	685
$\text{Al(OH)(H}_2\text{O)}_4^+ \cdot 13\text{H}_2\text{O}$	523	–	730
<i>cis</i> - $\text{Al(OH)}_2(\text{H}_2\text{O)}_4^+ \cdot 12\text{H}_2\text{O}$	441	–	626
<i>trans</i> - $\text{Al(OH)}_2(\text{H}_2\text{O)}_4^+ \cdot 12\text{H}_2\text{O}$	–	542	–
<i>cis</i> - $\text{Al(OH)}_2(\text{H}_2\text{O)}_3^+ \cdot 13\text{H}_2\text{O}$	503	–	659/745
<i>trans</i> - $\text{Al(OH)}_2(\text{H}_2\text{O)}_3^+ \cdot 13\text{H}_2\text{O}$	–	660	–
$\text{Al(OH)}_2(\text{H}_2\text{O)}_2^+ \cdot 14\text{H}_2\text{O}$	586	–	754

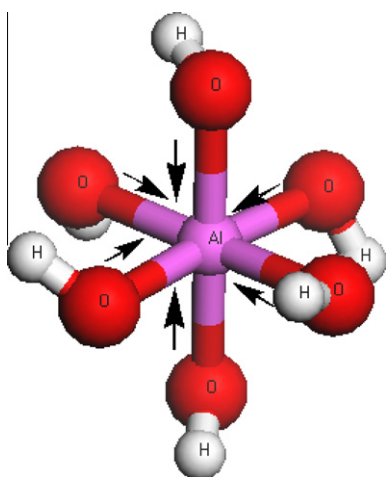


Fig. 8. The schematic diagram of Al–O vibration of Al(OH)_6^{3-} .

for Al(OH)^{2+} and Al(OH)_2^+ that can be observed in aqueous solution. The band at 490 cm^{-1} assigned to the Al–O vibration of $[\text{Al(OH)}_2\text{SO}_3]^+$ is observed in $\text{Al}_2(\text{SO}_4)_3$ solution, which verifies our point primarily [10b].

Meanwhile in aluminate solution, Al(OH)_4^- and $\text{Al}_2\text{O(OH)}_6^{2-}$ are identified [2d,2e]. However some species such as Al(OH)_6^{3-} is always mentioned and there is no unified point of view of its existences [1a,7,11a,11b]. Based on our calculation, the main Al–O Raman vibration of Al(OH)_6^{3-} is the symmetric stretching mode of AlO_6 . It is denoted as $\nu_s \text{ AlO}_6$, and presented in Fig. 8. The calculated frequency of $\nu_s \text{ AlO}_6$ is 510 cm^{-1} which is the same as the calculated Al–O–Al symmetric stretch frequency of $\text{Al}_2\text{O(OH)}_6^{2-}$ and is close to the observed data 535 cm^{-1} in aluminate solution. Therefore the existence of Al(OH)_6^{3-} should not be ruled out, which is consistent with the view of Walting [11d,11e].

3.2.2. The dimeric aquated aluminium species

The Raman spectra of $\text{Al}_2\text{O(OH)}_6^{2-}$ which is the identified component in aluminate solution has been studied for years. The main Al–O vibration modes of Raman spectra for $\text{Al}_2\text{O(OH)}_6^{2-}$ are depicted Fig. 9. The experimental Raman spectrum of $\text{Al}_2\text{O(OH)}_6^{2-}$ is characterized by two bands at 535 and 705 cm^{-1} [1a,7]. The calculated values 510 and 699 cm^{-1} are close to the observed ones and they can be assigned to the symmetric stretch of Al–O–Al ($\nu_s \text{ AlOAl}$) and the stretch of Al–OH ($\nu \text{ Al-OH}$) as depicted in Fig. 9 [2e].

4. Conclusions

Raman spectra of $\text{Al(H}_2\text{O)}_6^{3+}$ and Al(OH)_4^- species in GP, PCM, SM and SM-PCM models were calculated using DFT. The Al–O frequen-

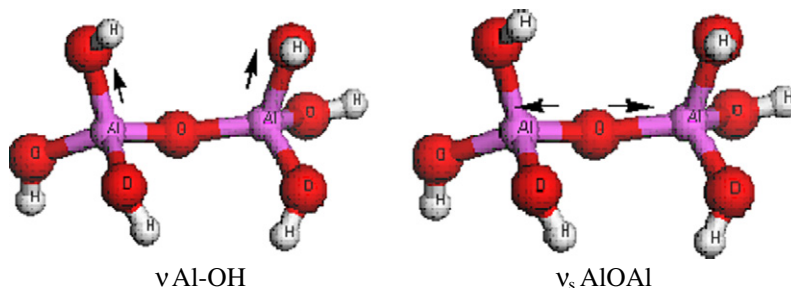


Fig. 9. The schematic diagrams of Al–O vibrations of $\text{Al}_2\text{O(OH)}_6^{2-}$.

cies of species in SM-PCM model correspond with the experimental frequencies. The main Al–O vibrations of $\text{Al}(\text{OH})_2^{2+}$, $\text{Al}(\text{OH})_2^+$, $\text{Al}(\text{OH})_6^{3-}$ and $\text{Al}_2\text{O}(\text{OH})_6^{2-}$ in Raman spectra were determined from the SM-PCM calculations. $\text{Al}(\text{OH})(\text{H}_2\text{O})_5^{2+}$ at 685 cm^{-1} , *trans*- $\text{Al}(\text{OH})_2(\text{H}_2\text{O})_4^+$ at 542 cm^{-1} , $\text{Al}(\text{OH})_6^{3-}$ at 510 cm^{-1} and $\text{Al}_2\text{O}(\text{OH})_6^{2-}$ at 510 and 699 cm^{-1} in Raman spectra would be observed in aqueous solution, respectively. Besides, according to our calculation, the controversial structure $\text{Al}(\text{OH})_6^{3-}$ cannot be excluded in aluminate solution. All these can give a guide for the analysis of the components in aqueous solutions of aluminium.

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Appendix A. Supplementary material

Supplementary data associated with this article can be found, in the online version, at doi:10.1016/j.molstruc.2010.07.012.

References

- [1] (a) P. Sipos, J. Mol. Liq. 146 (2009) 1;
(b) J.T. Gatlin, Z.L. Mensinger, L.N. Zakharov, D. Macinnes, D.W. Johnson, Inorg. Chem. 47 (2008) 1267;
(c) R.J. Evans, J.R. Rustad, W.H. Casey, J. Phys. Chem. A 112 (2008) 4125;
(d) Z.Y. Chen, Z.K. Luan, J.H. Fan, Z.G. Zhang, X.J. Peng, B. Fan, Colloid Surf. A 292 (2007) 110;
(e) W.H. Casey, Chem. Rev. 106 (2006) 1;
(f) J. Rowsell, L.F. Nazar, J. Am. Chem. Soc. 122 (2000) 3777.
- [2] (a) R.E. Connick, D.N. Fiat, J. Chem. Phys. 39 (1963) 1349;
(b) G. Johansson, Acta Chem. Scand. 16 (1962) 403;
(c) G. Johansson, Acta Chem. Scand. 14 (1960) 771;
(d) D.T. Richens, The Chemistry of Aqua Ions, Wiley, New York, 1997;
(e) R.J. Moolenaar, J.C. Evans, L.D. McKeever, J. Phys. Chem. 74 (1970) 3629.
- [3] (a) C.C. Perry, K.L. Shafran, J. Inorg. Biochem. 87 (2001) 115;
(b) A. Singhal, K.D. Keefer, J. Mater. Res. 9 (1994) 1973;
(c) T. Urabe, M. Tanaka, S. Kumakura, T. Tsugoshi, J. Mass Spectrom. 42 (2007) 591;
(d) A. Sarpola, V. Hietapelto, J. Jalonen, J. Jokela, R.S. Laitinen, J. Ramo, J. Mass Spectrom. 39 (2004) 1209.
- [4] (a) T.W. Swaddle, J. Rosenqvist, P. Yu, E. Bylaska, B.L. Phillips, W.H. Casey, Science 308 (2005) 1450;
(b) J.M. Ruiz, M.H. McAdon, J.M. Garcés, J. Phys. Chem. B 101 (1997) 1733;
(c) T. Ikeda, M. Hirata, T. Kimura, J. Chem. Phys. 124 (2006) 074503;
(d) H. Hanauer, R. Puchta, T. Clark, R. van Eldik, Inorg. Chem. 46 (2007) 1112.
- [5] (a) Z.S. Qian, H. Feng, W.J. Yang, Z.J. Zhang, Y.J. Wang, S.P. Bi, Dalton Trans. (2009) 1554;
(b) Z.S. Qian, H. Feng, Z.J. Zhang, W.J. Yang, J. Jin, Q. Miao, L. He, S.P. Bi, Dalton Trans. (2009) 521;
(c) Z.S. Qian, H. Feng, W.J. Yang, Q. Miao, L. He, S.P. Bi, Chem. Commun. (2008) 3930;
(d) Z.S. Qian, H. Feng, L.N. He, W.J. Yang, S.P. Bi, J. Phys. Chem. A 113 (2009) 5138;
(e) W.J. Yang, Z.S. Qian, Q. Miao, Y.J. Wang, S.P. Bi, Phys. Chem. Chem. Phys. 11 (2009) 2396.
- [6] (a) M. Kanzaki, J. Ceram. Soc. Jpn. 105 (1997) 91;
(b) J.D. Gale, A.L. Rohl, H.R. Watling, G.M. Parkinson, J. Phys. Chem. B 102 (1998) 10372.
- [7] (a) P. Sipos, G. Hefter, P.M. May, Talanta 70 (2006) 761;
(b) P. Sipos, P.M. May, G. Hefter, Dalton Trans. (2006) 368.
- [8] (a) J.R. Ferraro, K. Nakamoto, C.W. Brown, Introductory Raman Spectroscopy, vol. 1, Elsevier, Amsterdam, 2003;
(b) E. Smith, G. Dent, Modern Raman Spectroscopy – A Practical Approach, Wiley, New York, 2005.
- [9] (a) W.W. Rudolph, R. Mason, J. Solution Chem. 30 (2001) 527;
(b) W.W. Rudolph, R. Mason, C.C. Pye, Phys. Chem. Chem. Phys. 2 (2000) 5030.
- [10] (a) W. Rudolph, W.E. Steger, Z. Phys. Chem. 176 (1992) 185;
(b) W. Rudolph, S. Schönherr, Z. Phys. Chem. 172 (1991) 31;
(c) W. Rudolph, S. Schönherr, Z. Phys. Chem. 173 (1991) 167;
(d) W. Rudolph, S. Schönherr, Z. Phys. Chem. – Leipzig 270 (1989) 1121.
- [11] (a) Y. Chen, Q.M. Feng, K. Liu, Y.D. Chen, G.F. Zhang, Chem. Phys. Lett. 422 (2006) 406;
(b) J.R. Schoonover, S.L.L. Zhang, C.T. Johnston, J. Raman Spectrosc. 34 (2003) 404;
(c) C.T. Johnston, S.F. Agnew, J.R. Schoonover, J.W. Kennedy, B. Page, J. Osborn, R. Corbin, Environ. Sci. Technol. 36 (2002) 2451;
(d) H. Watling, Appl. Spectrosc. 52 (1998) 250;
(e) H.R. Watling, S.D. Fleming, W. van Bronswijk, A.L. Rohl, J. Chem. Soc. Dalton Trans. 23 (1998) 3911;
(f) L.A. Carreira, V.A. Maroni, J.W. Swaine, R.C. Plumb, J. Chem. Phys. 45 (1966) 2216;
(g) E.R. Lippincott, J.A. Psellos, M.C. Tobin, J. Chem. Phys. 20 (1952) 536.
- [12] (a) T.M.C. Faro, G.P. Thim, M.S. Skaf, J. Chem. Phys. 132 (2010) 114509;
(b) J.A. Tossell, Am. Mineral. 84 (1999) 1641.
- [13] (a) L. Guimarães, H.A. de Abreu, H.A. Duarte, Chem. Phys. 333 (2007) 10;
(b) H.A. de Abreu, L. Guimarães, H.A. Duarte, J. Phys. Chem. A 110 (2006) 7713;
(c) F.P. Rotzinger, Chem. Eur. J. 13 (2007) 800;
(d) S. Miertus, E. Scrocco, J. Tomasi, Chem. Phys. 55 (1981) 117.
- [14] (a) C.W. Bock, G.D. Markham, A.K. Katz, J.P. Glusker, Inorg. Chem. 42 (2003) 1538;
(b) A.J. Sillanpää, J.T. Paivarinta, M.J. Hotokka, J.B. Rosenholm, K.E. Laasonen, J. Phys. Chem. A 105 (2001) 10111.
- [15] (a) Z.S. Qian, H. Feng, W.J. Yang, X.Y. Jin, S.P. Bi, Dalton Trans. (2009) 8013;
(b) Z.S. Qian, H. Feng, Z.J. Zhang, W.J. Yang, M. Wang, Y.J. Wang, S.P. Bi, Geochim. Cosmochim. Acta 73 (2009) 1588;
(c) Z.S. Qian, H. Feng, W.J. Yang, S.P. Bi, J. Am. Chem. Soc. 130 (2008) 14402.
- [16] (a) C.W. Bock, G.D. Markham, A.K. Katz, J.P. Glusker, Theor. Chem. Acc. 115 (2006) 100;
(b) G.D. Markham, C.W. Bock, J.P. Glusker, J. Phys. Chem. B 106 (2002) 5118.
- [17] M.J. Frisch, G.W. Trucks, H.B. Schlegel, G.E. Scuseria, M.A. Robb, J.R. Cheeseman, J.A. Montgomery Jr., T. Vreven, K.N. Kudin, J.C. Burant, J.M. Millam, S.S. Iyengar, J. Tomasi, V. Barone, B. Mennucci, M. Cossi, G. Scalmani, N. Rega, G.A. Petersson, H. Nakatsuji, M. Hada, M. Ehara, K. Toyota, R. Fukuda, J. Hasegawa, M. Ishida, T. Nakajima, Y. Honda, O. Kitao, H. Nakai, M. Klene, X. Li, J.E. Knox, H.P. Hratchian, J.B. Cross, V. Bakken, C. Adamo, J. Jaramillo, R. Gomperts, R.E. Stratmann, O. Yazyev, A.J. Austin, R. Cammi, C. Pomelli, J. Ochterski, P.Y. Ayala, K. Morokuma, G.A. Voth, P. Salvador, J.J. Dannenberg, V.G. Zakrzewski, S. Dapprich, A.D. Daniels, M.C. Strain, O. Farkas, D.K. Malick, A.D. Rabuck, K. Raghavachari, J.B. Foresman, J.V. Ortiz, Q. Cui, A.G. Baboul, S. Clifford, J. Cioslowski, B.B. Stefanov, G. Liu, A. Liashenko, P. Piskorz, I. Komaromi, R.L. Martin, D.J. Fox, T. Keith, M.A. Al-Laham, C.Y. Peng, A. Nanayakkara, M. Challacombe, P.M.W. Gill, B.G. Johnson, W. Chen, M.W. Wong, C. Gonzalez, J.A. Pople, Gaussian 03, Revision D.01, Gaussian, Inc., Wallingford CT, 2004.
- [18] (a) W. Koch, M.C. Holthausen, A Chemist's Guide to Density Functional Theory, second ed., Wiley, VCH Verlag, GmbH, 2001 (Copyright ©);
(b) C.Y. Lee, W. Yang, R.G. Parr, Phys. Rev. B 37 (1988) 785.
- [19] (a) Q. Miao, Q. Cao, S.P. Bi, J. Chem. Phys. 121 (2004) 4650;
(b) A.E. Reed, R.B. Weinstock, F. Weinhold, J. Chem. Phys. 83 (1985) 735.
- [20] (a) L.H.V. Lim, T.S. Hofer, A.B. Pribil, B.M. Rode, J. Phys. Chem. B 113 (2009) 4372;
(b) M. Sandstrom, I. Persson, F. Jalilehvand, P. Lindquist-Reis, D. Spangberg, K. Hermansson, J. Synchrotron Radiat. 8 (2001) 657;
(c) M.B. Hay, S.C.B. Myneni, J. Phys. Chem. A 112 (2008) 10595.
- [21] M.A. Marques, M.A.S. Oliveira, J.R. Rodrigues, J. Chem. Soc. Faraday Trans. 86 (1990) 471.
- [22] (a) I. Diaz-Acosta, J. Baker, W. Cordes, P. Pulay, J. Phys. Chem. A 105 (2001) 238;
(b) J. Baker, A.A. Jarzecki, P. Pulay, J. Phys. Chem. A 102 (1998) 1412;
(c) G. Rauhut, P. Pulay, J. Am. Chem. Soc. 117 (1995) 4167.
- [23] (a) D.A. Rozmanov, O.V. Sizova, M.Y. Skripkin, K.A. Burkov, Spectrochim. Acta A 62 (2005) 92;
(b) W.W. Rudolph, D. Fischer, M.R. Tomney, C.C. Pye, Phys. Chem. Chem. Phys. 6 (2004) 5145;
(c) A.A. Jarzecki, A.D. Anbar, T.G. Spiro, J. Phys. Chem. A 108 (2004) 2726;
(d) W.W. Rudolph, C.C. Pye, Phys. Chem. Chem. Phys. 4 (2002) 4319;
(e) W.W. Rudolph, C.C. Pye, G. Irmer, J. Raman Spectrosc. 33 (2002) 177;
(f) C.C. Pye, W.W. Rudolph, J. Phys. Chem. A 102 (1998) 9933;
(g) C.C. Pye, W.W. Rudolph, R.A. Poirier, J. Phys. Chem. 100 (1996) 601.
- [24] J. Mink, C. Németh, L. Hajba, A. Sandström, P.L. Goggins, J. Mol. Struct. 661 (2003) 141.
- [25] A. Serr, R.R. Netz, Int. J. Quantum Chem. 106 (2006) 2960.
- [26] J. Neugebauer, M. Reiher, C. Kind, B.A. Hess, J. Comput. Chem. 23 (2002) 895.
- [27] (a) H. Cybulski, J. Sadlej, Chem. Phys. 342 (2007) 163;
(b) A. Lenz, L. Ojamäe, Chem. Phys. Lett. 418 (2006) 361;
(c) J. Sadlej, Chem. Phys. Lett. 333 (2001) 485;
(d) T. Uchino, T. Yoko, Phys. Chem. Chem. Phys. 1 (1999) 3473.